MiSiCNet: Minimum Simplex Convolutional Network for Deep Hyperspectral Unmixing

Behnood Rasti, *Senior Member, IEEE*, Bikram Koirala, *Member, IEEE*, Paul Scheunders, *Senior Member, IEEE*, and Jocelyn Chanussot, *Fellow, IEEE*

***Abstract*— In this article, we propose a minimum simplex convolutional network (MiSiCNet) for deep hyperspectral unmixing. Unlike all the deep learning-based unmixing methods proposed in the literature, the proposed convolutional encoder–decoder architecture incorporates spatial information and geometrical information of the hyperspectral data in addition to the spectral information. The spatial information is incorporated using convolutional filters and implicitly applying a prior on the abundances. The geometrical information is exploited by incorporating a minimum simplex volume penalty term in the loss function for the endmember estimation. This term is beneficial when there are no pure material pixels in the data, which is often the case in real-world applications. We generated simulated datasets, where we consider two different no-pure pixel scenarios. In the first scenario, there are no pure pixels but at least two pixels on each facet of the data simplex (i.e., mixtures of two pure materials). The second scenario is a complex case with no pure pixels and only one pixel on each facet of the data simplex. In addition, we evaluate the performance of MiSiCNet in three real datasets. The experimental results confirm the robustness of the proposed method to both noise and the absence of pure pixels. In addition, MiSiCNet considerably outperforms the state-of-the-art unmixing approaches. The results are given in terms of spectral angle distance in degree for the endmember estimation and the root mean square error in percentage for the abundance estimation. MiSiCNet was implemented in Python (3.8) using PyTorch as the platform for the deep network and is available online: https://github.com/BehnoodRasti/MiSiCNet.**

***Index Terms*— Blind unmixing, convolutional neural network, deep learning, deep prior, endmember extraction, hyperspectral image, minimum simplex volume, unmixing.**

I. Introduction

A SPECTRAL pixel is generally a mixture of the pure spectra of the materials within the pixel, called endmembers [1], [2]. Estimating the fractional abundances of the endmembers within spectral pixels is called spectral unmixing. If the materials are intimately mixed, and multiple reflections are received at the sensor, then nonlinear unmixing is utilized. On the other hand, linear unmixing is found helpful in remote sensing applications, where hyperspectral images often consist of homogeneous regions of single materials [3], [4].

If there exist pure pixels for each material within the scene, then the endmembers can be easily extracted using a geometrical approach relying on the simplex of the data. After extracting the endmembers, the abundances can be estimated by minimizing the least-squared errors between the actual spectra and reconstructed spectra from the endmembers subjected to the physical constraints on the abundances, i.e., the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC) [5].

The endmembers are assumed to be located at the vertices of the data simplex. Therefore, they can be extracted by maximizing the data simplex, e.g., by simplex volume maximization (SiVM) [6]. In [7], convex optimizations were used to solve the maximum volume inscribed simplex (MVIS) problem. MVIS searches for the data points whose simplex convex hull is inscribed in the data convex hull, while the simplex volume is maximized. Alternatively, the vertices can be selected as the extreme points after iteratively projecting the data onto a particular direction. For instance, the vertex component analysis (VCA) [8] selects endmembers iteratively by projecting the data into an orthogonal direction to the subspace spanned by the already selected endmember. The pixel purity index (PPI) [9] scores the spectral vectors by projecting them onto a large set of random vectors (called skewers) and counting the number of times that each vector is an extreme point. N-FINDR [10] searches for pure pixels that form the largest simplex by gradually inflating a simplex inside the data.

In the absence of pure pixels, the geometrical approaches can still be successful when enough data points are available on the facets of the data simplex. Then, virtual endmembers located at the vertices can be estimated. For instance, the minimum volume enclosing simplex (MVES) problem seeks the minimum simplex that encloses the data points [11], [12] and, therefore, does not rely on the pure pixel assumption. In [13], a maximum volume inscribed ellipsoid (MVIE) method was proposed. MVIE seeks the maximum volume ellipsoid contained within the convex hull of the data points. This volume is maximal when touching the facets of the data simplex. The contact points provide clues on how to estimate the virtual endmembers.

Manuscript received November 9, 2021; revised December 13, 2021; accepted January 23, 2022. Date of publication January 27, 2022; date of current version March 17, 2022. The work of Behnood Rasti was supported by the Alexander-von-Humboldt-Stiftung/Foundation. The work of Bikram Koirala was supported by the Belgian Science Policy Office (BELSPO) in the frame of the STEREO III Programme under Project GEOMIX (SR/06/357). *(Corresponding author: Behnood Rasti.)*

Behnood Rasti is with the Machine Learning Group, Helmholtz Institute Freiberg for Resource Technology, Helmholtz-Zentrum Dresden-Rossendorf, 09599 Freiberg, Germany (e-mail: b.rasti@hzdr.de).

Bikram Koirala and Paul Scheunders are with the imec-Vision Lab, University of Antwerp (CDE), B-2610 Antwerp, Belgium (e-mail: bikram.koirala@uantwerpen.be; paul.scheunders@uantwerpen.be).

Jocelyn Chanussot is with the Laboratoire Jean Kuntzmann (LJK), Centre National de la Recherche Scientifique [French National Centre for Scientific Research (CNRS)], Inria, Grenoble Institute of Technology (Grenoble INP), Université Grenoble Alpes, F-38000 Grenoble, France (e-mail: jocelyn.chanussot@grenoble-inp.fr).

Digital Object Identifier 10.1109/TGRS.2022.3146904

1558-0644 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

In no pure pixel scenarios, estimating the endmembers and the abundances simultaneously is called blind unmixing. A common procedure is to use constrained (penalized) least-squared error minimization (also referred to as nonnegative matrix factorization in the literature [14]) to form an optimization problem with respect to both endmembers and abundances [15], [16]. In those problems, the geometrical information can be induced by using a minimum volume-based penalty term. In [17], the square of the simplex volume was selected as the penalty term in a fully constrained penalized least-squares estimation. The Euclidean distances between the endmembers and the center of the data points are considered in [14] as a geometrical penalty. Li *et al.* [18] proposed a geometrical penalty defined by the Euclidean distance between the endmembers and the extremes of the data simplex extracted by a geometrical approach (i.e., VCA). In [19], the total variation (TV) of all the endmembers is proposed as a penalty term. In these optimization problems, the selection of the regularization parameter, which is the tradeoff between the fidelity term and the penalty term, is often a complex problem. In [20], a parameter selection technique is proposed for the fully constrained penalized least-squares method with the three different geometrical penalties proposed in [14], [18], and [19].

Alternatively, the unmixing problem can be reformulated into a nonconvex minimization (or a nonconcave maximization) problem, which seeks the minimum volume data simplex, subjected to the ASC and ANC [21], [22]. Note that the initialization can considerably affect the performance of the aforementioned nonconvex approaches.

Sparse unmixing is another group of unmixing techniques that estimate the fractional abundances using sparse regression techniques, relying on a rich and well-designed library of pure spectra. Each spectrum is assumed to be a sparse linear combination of the dictionary elements, i.e., the library. The optimization problem is often defined in the form of penalized least squares with a sparsity-promoting penalty applied on the abundances [23], [24]. Recently, in [25], a deep learning approach was proposed for sparse unmixing using a convolutional neural network (SUnCNN). The major disadvantage of sparse unmixing techniques is their dependence on the spectral library, which can considerably affect the abundance estimation performance. On the other hand, a specific group of sparse unmixing approaches considers the endmembers' spectral variability by using a dictionary of endmember bundles generated from the data itself. An example of such approach is the Collaborative Least Absolute Shrinkage and Selection Operator (LASSO [26]) [27] [28].

In recent years, deep learning techniques attracted increasing attention in remote sensing, including hyperspectral image analysis. A variety of architectures has been developed and proposed for different applications, such as hyperspectral image classification [29]. For deep hyperspectral unmixing, the most widely used architecture is based on autoencoders. The abundances are often generated by enforcing the constraints (i.e., ANC and ASC) in the final layer of the encoder. The decoder has one layer that reconstructs the signal, and its weights are the endmembers. In [30], a stack of nonnegative sparse autoencoders (SNSA) was proposed in which the last autoencoder is utilized for unmixing, and the rest of the network improves the robustness with respect to outliers. The deep autoencoder network (DAEN) [31] exploits a stacked autoencoder to initialize a variational autoencoder to estimate the endmembers and the abundances. In [32], a deep network was proposed, which uses a variational autoencoder to generate the endmembers. An untied Denoising Autoencoder with Sparsity (uDAS) was proposed in [33] for spectral unmixing. uDAS benefits from an additional denoising constraint applied to the decoder and an $\ell_{2,1}$ sparsity constraint applied to the decoder. A sparse autoencoder was proposed in [34], which uses a loss function with a Kullback–Leibler divergence term, an SAD similarity, and a sparsity term. We should note that, in all the abovementioned autoencoder-based techniques, the spatial information is ignored, as, for training the network, the autoencoder receives a spectral pixel at a time [35]. In [36], an unmixing technique based on an autoencoder network is proposed, which incorporates the spatial correlation between pixels by utilizing an adaptive abundance smoothing method. In [37], patchwise or cubewise convolutional autoencoders were proposed to incorporate the spatial information. In [38], parallel autoencoders were applied to spectral patches to exploit the spatial information. Recently, a 3-D convolutional autoencoder was proposed for supervised hyperspectral unmixing (i.e., the endmembers are assumed to be known) in [39]. The patchwise and cubewise approaches that exploit the convolutional filters are beneficial for endmember estimation since they capture the variability of the spectra. On the other hand, they may blur the abundance maps [40]. Unmixing using deep image prior (UnDIP) [41] utilizes a convolutional encoder–decoder architecture and a deep image prior (DIP) [42], [43]. UnDIP is robust to noise and provides sharp abundances. However, it relies on a geometrical approach (i.e., SiVM) to extract the endmembers, i.e., it only considers pure pixel scenarios. The cycle-consistency unmixing network (CyCUNet) [44] utilizes two convolutional autoencoders that are cascaded and performed cyclically. The proposed loss function contains two terms for spectral reconstruction and one for abundance reconstruction to incorporate high-level semantic information.

A major problem of the DL-based approaches is the absence of geometrical information. Therefore, as we will show in Section IV, most of them fail to accurately estimate the endmembers when there are no pure pixels in the dataset. On the other hand, the advantage of simplex volume minimization for blind unmixing in no pure pixel scenarios has been proven [45]. In this article, we propose a convolutional encoder–decoder architecture for blind spectral unmixing called minimum simplex convolutional network (MiSiCNet). MiSiCNet utilizes a deep encoder–decoder network that incorporates both spatial information and geometrical information. The spatial information is incorporated by using the convolutional operator and implicitly applying a regularizer on the abundances. The geometrical information is exploited using a simplex volume penalized loss function. We show that MiSiCNet is superior in unmixing datasets that do not contain pure pixels. MiSiCNet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

was implemented in Python (3.8) using PyTorch as the platform for the deep network and is available online: https://github.com/BehnoodRasti/MiSiCNet.

A. Contributions and Novelties

The contribution of the proposed method is threefold.

1) MiSiCNet is the first deep blind unmixing network that considers the geometry of the data points by employing a simplex volume penalized loss function to incorporate the geometrical information, while the DL-based unmixing techniques from the literature directly optimize either reconstruction errors or spectral angle distance (SAD). Some conventional techniques, such as [17] and [20], incorporate the geometrical information using a simplex volume penalized least-squares optimization. However, they do not incorporate spatial information, mostly because the optimization problem becomes too complex, and the selection of the tuning parameter is complicated. In the experiments, we show that incorporating the geometrical information is crucial for the endmember estimation when there are no pure pixels available.
2) We propose a deep unsupervised convolutional network that solves blind unmixing problems with regularizers on both abundances and endmembers. In other words, we demonstrate how to turn a blind unmixing problem into the optimization of a deep unsupervised convolutional network. In addition, we show how to implement all the physical constraints via applying constraints on the network parameters.
3) We propose a convolutional encoder–decoder with a skip connection that incorporates the spatial information using both convolutional filters and an implicitly induced regularizer on the abundances. In addition, the skip connection augments the low-level features to high-level features, which sharpens the abundance maps.

The remaining of this article is organized as follows. The proposed unmixing methodology is explained in detail in Section II. The experimental results are shown and discussed in Section IV. Section V concludes this article.

II. Methodology

A. Hyperspectral Image Modeling

We assume that the mixing model for the observed HSI is given by

$$\mathbf{Y} = \mathbf{EA} + \mathbf{N}, \quad \text{s.t. } \mathbf{A} \geq 0, \quad \mathbf{1}_r^T \mathbf{A} = \mathbf{1}_n^T, \quad 0 \leq \mathbf{E} \leq 1 \tag{1}$$

where $\mathbf{Y} \in \mathbb{R}^{p \times n}$ is the observed HSI, with n pixels and p bands, $\mathbf{N} \in \mathbb{R}^{p \times n}$ is the noise, $\mathbf{E} \in \mathbb{R}^{p \times r}$, and $\mathbf{A} \in \mathbb{R}^{r \times n}$, $r \ll p$, contain the r endmembers and their fractional abundances, respectively. $\mathbf{1}_n$ indicates an n-component column vector of ones. In blind unmixing scenarios, the task is to estimate both $\mathbf{E}$ and $\mathbf{A}$ simultaneously.

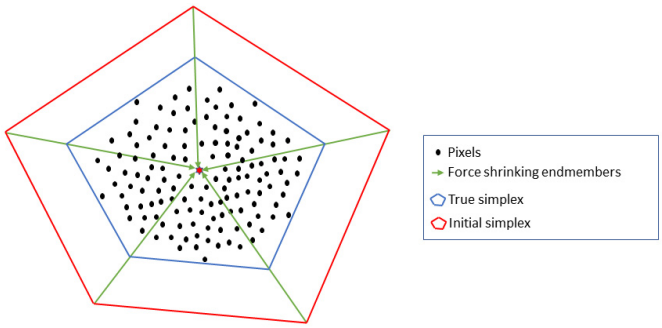


Fig. 1. Geometrical illustration of the minimum volume constraint, pulling the endmembers toward the center of the data simplex.

A general optimization problem for estimating both $\mathbf{E}$ and $\mathbf{A}$ is given by

$$\begin{aligned}(\hat{\mathbf{A}}, \hat{\mathbf{E}}) = \arg\min_{\mathbf{A},\mathbf{E}} \frac{1}{2}\|\mathbf{Y} - \mathbf{EA}\|_F^2 + \lambda\phi(\mathbf{E}) + \beta R(\mathbf{A}) \\ \text{s.t. } \mathbf{A} \geq 0, \mathbf{1}_r^T\mathbf{A} = \mathbf{1}_n^T, 0 \leq \mathbf{E} \leq 1\end{aligned} \tag{2}$$

where the first term is the fidelity term, ϕ and R are penalty terms on endmembers and abundances, respectively, and λ and β control the tradeoff between the penalty terms and the fidelity term. $\|.\|_F$ denotes the Frobenius norm. The problem is solved, subjected to the nonnegativity and sum to one constraint. The penalty functions are often chosen based on the prior knowledge of the abundances and endmembers. The function R is often selected to capture the spatial correlation in the dataset. For instance, a TV penalty or a sparsity enforcing penalty is often a common choice for R since it induces piecewise smoothness on the abundance maps. The function ϕ is often chosen to minimize the volume of the data simplex. A reasonable choice is to enforce the endmembers toward the center of the data simplex [14]

$$\phi(\mathbf{E}) = \|\mathbf{E} - \mathbf{m}\mathbf{1}_\mathbf{r}^\mathbf{T}\|_F^\mathbf{2} \tag{3}$$

where $\mathbf{m}$ is a vector that contains the mean values of the spectral pixels. A geometrical interpretation of this term is given in Fig. 1. As can be seen, by minimizing $\phi(\mathbf{E})$, the endmembers are pulled toward the vertices of the true data simplex from their initial values, i.e., the vertices of the initial data simplex.

Therefore, the problem to solve becomes

$$\begin{aligned}(\hat{\mathbf{A}}, \hat{\mathbf{E}}) = \arg\min_{\mathbf{A},\mathbf{E}} \frac{1}{2}\|\mathbf{Y} - \mathbf{EA}\|_F^2 + \lambda\|\mathbf{E} - \mathbf{m}\mathbf{1}_\mathbf{r}^\mathbf{T}\|_F^\mathbf{2} \\ +\beta R(\mathbf{A}) \text{ s.t. } \mathbf{A} \geq 0, \quad \mathbf{1}_r^T\mathbf{A} = \mathbf{1}_n^T \quad , 0 \leq \mathbf{E} \leq 1.\end{aligned} \tag{4}$$

For more discussion on the selection of ϕ, we refer to [17]. In [14], function R is selected to promote sparsity on $\mathbf{A}$. It is worth mentioning that selecting λ and β is not a trivial task. This is partially addressed in [17] by neglecting the spatial dependence of the abundance maps, i.e., $\beta = 0$.

B. Deep Unmixing Using MiSiCNet

A general inverse image reconstruction task can be formulated as an optimization problem

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{X}\|_F^2 + \lambda R(\mathbf{X}) \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

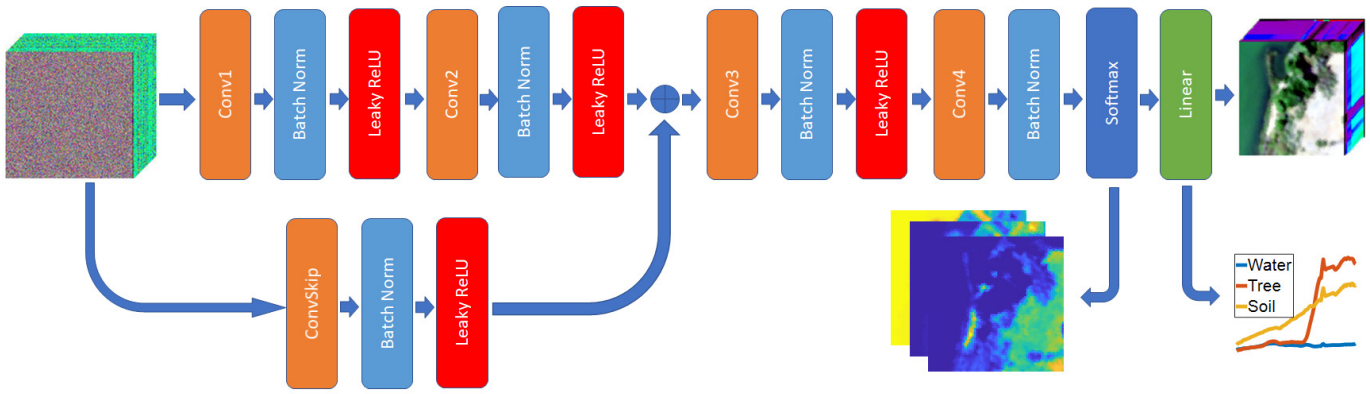


Fig. 2. Architecture of MiSiCNet. The proposed convolutional network architecture uses a skip connection and five convolutional layers. The softmax layer is used as the final activation function, followed by a linear layer that generates the abundances and endmembers, respectively.

where λ is the tuning parameter that acts as a tradeoff between the fidelity term and the regularizer (R). The regularizer selection depends on the application and the available prior knowledge. On the other hand, DIP [42], [43] shows that the selection of a suitable regularizer can be substituted by optimizing a deep network's parameters (i.e., weights and biases)

$$\hat{\theta} = \arg\min_{\boldsymbol{\theta}} \frac{1}{2}\|\mathbf{Y} - f_{\theta}(\mathbf{Z})\|_F^2 \text{ s.t. } \hat{\mathbf{X}} = f_{\hat{\theta}}(\mathbf{Z}) \tag{6}$$

where $\mathbf{Z}$ is the network input that is fixed throughout the optimization and is often selected as a random input image. Here, we explain how we solve (4) using a deep neural network and (6).

Deep unmixing techniques utilize deep encoder–decoder architectures to estimate abundances and endmembers. The encoder part often contains several layers to estimate the abundances at the bottleneck, and the decoder contains only one linear layer. The reconstruction loss is often minimized to train the network. Let us denote the encoder network with f_{θ_1} with network parameters (i.e., weights and biases) denoted by θ_1. The decoder part is only a linear layer that we simply show by a set of weights i.e., θ_2. Momentarily, neglecting all the constraints, using (6), we can turn the optimization problem (4) into an optimization of the network parameters

$$\begin{aligned}(\hat{\theta}_1, \hat{\mathbf{E}}) = \arg\min_{\theta_1, \mathbf{E}} \frac{1}{2}\|\mathbf{Y} - \mathbf{E}f_{\theta_1}(\mathbf{Z})\|_F^2 + \lambda\|\mathbf{E} - \mathbf{m1_r^T}\|_F^2 \\ \text{s.t. } \hat{\mathbf{A}} = f_{\hat{\theta}_1}(\mathbf{Z}).\end{aligned} \tag{7}$$

As $\mathbf{E}$ contains the weights of the final linear layer (i.e., the decoder), we rewrite (7) as

$$\begin{aligned}(\hat{\theta}_1, \hat{\theta}_2) = \arg\min_{\theta_1, \theta_2} \frac{1}{2}\|\mathbf{Y} - \theta_2 f_{\theta_1}(\mathbf{Z})\|_F^2 + \lambda\|\theta_2 - \mathbf{m1}_r^T\|_F^2 \\ \text{s.t. } \hat{\mathbf{E}}\hat{\mathbf{A}} = \hat{\theta}_2 f_{\hat{\theta}_1}(\mathbf{Z})\end{aligned} \tag{8}$$

where $\hat{\mathbf{Y}} = \hat{\mathbf{E}}\hat{\mathbf{A}}$. Hence, the optimization problem (4) can be solved using a deep network with a loss function given by

$$\mathcal{L}(\mathbf{Y}, \hat{\mathbf{Y}}, \hat{\theta}_2, \mathbf{m}) = \frac{1}{2}\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F^2 + \lambda\|\hat{\theta}_2 - \mathbf{m1}_r^T\|_F^2. \tag{9}$$

To enforce both the ASC and the ANC, we use a softmax function in the final layer of the encoder

$$\text{softmax}(\mathbf{A}) = \frac{e^{\mathbf{A}_{ij}}}{\sum_{i=1}^{r} e^{\mathbf{A}_{ij}}} \quad \forall i, j \tag{10}$$

and the weights of the decoder (i.e., θ_2) are constrained between 0 and 1 in every step of the optimization to enforce the endmember constraint i.e., $0 \leq \mathbf{E} \leq 1$.

III. Architecture of MiSiCNet

The architecture of MiSiCNet is visualized in Fig. 2. The main architecture is based on the convolutional encoder–decoder with a skip connection. The skip connection is able to learn the identity function when the parameters become zero and avoids vanishing gradients in a deep network. In addition, via the skip connection, the information is shifted from low-level features to high-level (deep) features, which considerably helps to keep details such as the structures in the deep feature representation. We use four convolutional layers (Conv), excluding the skip connection (ConvSkip). To preserve the abundances' spatial resolution, we do not apply any downsampling and use reflection padding in each layer. The number of filters for each convolutional layer is given in Table I. Each convolutional layer is followed by a batch normalization (BN) layer that speeds up the learning process and provides more robustness for selecting the hyperparameters. To promote the nonlinearity, we use a leaky rectified linear unit (ReLU [46]) as the nonlinear activation function for all the convolutional layers, except for the last one where we use the softmax to enforce the ASC and ANC. We select random noise with the same size as the dataset as the input ($\mathbf{Z}$) and train the network iteratively to map the input to the dataset.

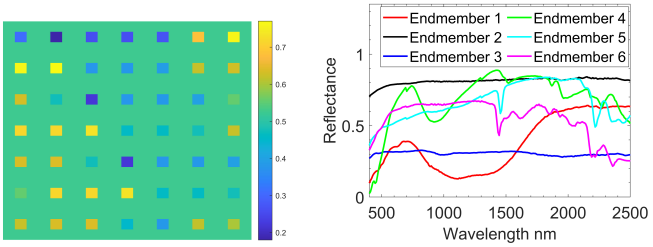
A. Hyperparameter Selection

The selection of hyperparameters is always challenging in a deep network. In this work, we fix all the hyperparameters used in the experiments, except the tuning parameter (λ). We set the negative slope of Leaky ReLU to 0.1. The convolutional layers use kernels of size 3×3, except for ConvSkip, which uses a kernel of size 1×1. We use four filters in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HYPERPARAMETERS OF MiSiCNet USED IN THE EXPERIMENTS

Hyperparameters				
	Input Ch.	Ouput Ch.	Filter Size	Stride
Conv1	p	256	3x3	1
Conv2	256	256	3x3	1
Conv3	260	256	3x3	1
Conv4	256	r	3x3	1
ConvSkip	p	4	1x1	1
		Negative Slope		
Leaky ReLU		0.1		
	Type	Learning Rate		Iterations
Optimizer	Adam	0.001		8000



(a) (b)

Fig. 3. Simulated Dataset 1. (a) Band number 70 (1050 nm). (b) Endmembers.

skip connection, 256 filters for Conv1, Conv2, and Conv3, and r (the number of pure materials) filters for Conv4. We should note that MiSiCNet is an unsupervised network, and we train the network over the entire dataset iteratively. Therefore, the number of iterations is a hyperparameter for the proposed network. We set the number of iterations to 8000. We also use exponentially weighted averaging over the outputs to make the technique robust to a possible jump in the loss function at the final iteration. Finally, we use the Adam optimizer and set the learning rate (LR) to 0.001. We use the PyTorch deep learning platform for network implementation. Table I lists the hyperparameters used in the experiments.

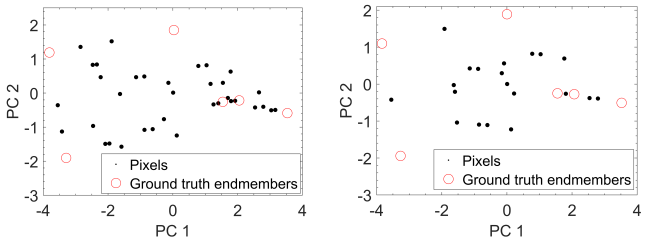
IV. Experimental Results

We performed experiments on two simulated datasets and three real datasets. The description of the datasets is given in the following.

A. Hyperspectral Data Description

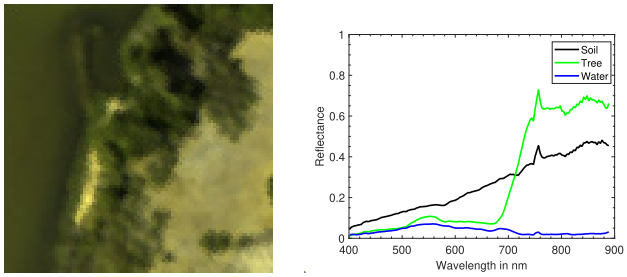
1) Simulated Dataset 1: A dataset of 105 × 105 pixels [see Fig. 3(a)] is simulated by generating linear mixtures of six endmembers. These endmembers are shown in Fig. 3(b) and contain 224 reflection values in the wavelength range [400–2500] nm. A PCA reduced data manifold is shown in Fig. 4(a). As can be observed, no pure pixels (red circles) are available in the dataset, but at least two mixed data points are available on each facet to geometrically reconstruct virtual endmembers.

2) Simulated Dataset 2: The second simulated dataset contains 105 × 105 pixels that are simulated by the linear combination of six endmembers (see Fig. 3(b) for endmembers).



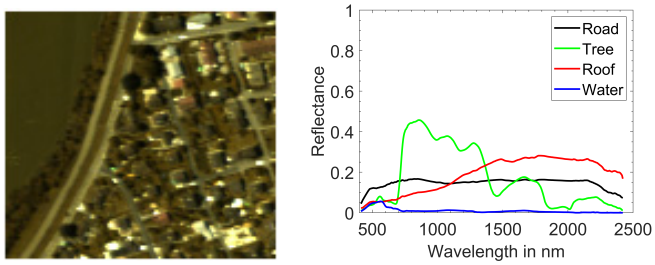
(a) (b)

Fig. 4. PCA reduced data manifolds. (a) Simulated dataset 1. (b) Simulated dataset 2.



(a) (b)

Fig. 5. Samson image. (a) True-color image (red: 571.01 nm, green: 539.53 nm, and blue: 432.48 nm). (b) Endmembers.



(a) (b)

Fig. 6. Apex image. (a) True-color image (red: 572.2 nm, green: 532.3 nm, and blue: 426.5 nm). (b) Endmembers.

The main difference with the simulated dataset 1 is that each facet of the data manifold contains only one mixed data point [see Fig. 4(b)], making it challenging to reconstruct virtual endmembers geometrically.

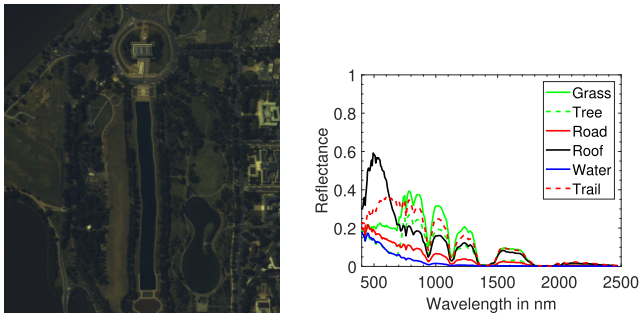
3) Samson: The Samson hyperspectral dataset ([47]) [see Fig. 5(a)] contains 95 × 95 pixels, having 156 bands in the wavelength range from 401 to 889 nm. There are three main materials (i.e., soil, tree, and water). The ground-truth endmembers shown in Fig. 5(b) were manually selected from the hyperspectral image, and the ground-truth fractional abundances were generated using FCLSU.

4) Apex: Fig. 6(a) shows a cropped image of the Apex dataset [48], as used in this article. It contains 111 × 122 pixels and 285 bands covering the wavelength range from 413 to 2420 nm. Four ground-truth endmembers (i.e., water, tree, road, and roof), as shown in Fig. 6(b), were manually selected from the hyperspectral image, and the ground-truth fractional abundances were generated using FCLSU.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RMSE (SIMULATED 1). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
20dB	11.13±1.93	8.75±0.56	9.40±1.73	17.97±2.93	10.69±1.53	**3.81±0.20**	3.90±0.08
30dB	12.28±3.44	7.19±0.87	9.40±1.91	17.83±4.73	11.22±2.29	1.81±0.42	**1.80±0.04**
40dB	12.29±2.49	7.73±0.87	9.36±2.03	17.97±7.02	10.07±1.68	1.70±0.90	**1.23±0.05**
50dB	11.15±2.72	7.36±0.88	8.46±2.04	16.07±0.76	10.35±2.02	3.71±0.74	**1.21±0.08**



(a) (b)

Fig. 7. Washington DC Mall image. (a) True-color image (red: 572.7 nm, green: 530.1 nm, and blue: 425.0 nm). (b) Endmembers.

5) Washington DC Mall: Washington DC Mall is an airborne hyperspectral dataset captured over the Washington DC Mall using the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor.[1] Fig. 7(a) shows the cropped data used in this article, which contains 319 × 292 pixels in 191 bands over the spectral range from 400 to 2400 nm. The ground-truth endmembers for six classes, i.e., grass, tree, roof, road, water, and trail, were manually selected from the hyperspectral image [see Fig. 7(b)], and FCLSU was used to estimate the ground-truth fractional abundances.

B. Experimental Setup

Six unmixing techniques from different unmixing categories were used as competing methods in the experiments.

1) *Geometrical Unmixing:* FCLSU [5] using VCA [8] for endmember extraction.
2) *Geometrical and Blind Unmixing:* NMF-QMV [20].
3) *Sparse Unmixing:* Collaborative LASSO (Collab) [28].
4) *Deep Unmixing Methods:* uDAS [33], UnDIP [41], and CyCUNet [44].

For MiSiCNet, we select all the hyperparameters, as discussed in Section III-A, and we set λ to 0.1, 0.3, and 100 for the simulated dataset 1, the simulated dataset 2, and real datasets, respectively. We choose all the parameters for the competing methods according to the reported default values. Note that, for NMF-QMV, we choose the "center" as the optional penalty term to have a fair comparison with MiSiCNet.

Quantitative results are provided by the root mean square error (RMSE) in percentage between the estimated and ground-truth abundance fractions

$$\mathrm{RMSE}\big(\hat{\mathbf{A}}, \mathbf{A}\big) = 100 \times \sqrt{\frac{1}{rn}\sum_{i=1}^{r}\sum_{j=1}^{n}\big(\hat{\mathbf{A}}_{ij} - \mathbf{A}_{ij}\big)^2} \quad (11)$$

[1]https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html

and the SAD in degree between the estimated and ground-truth endmembers

$$\mathrm{SAD}\big(\mathbf{E}, \hat{\mathbf{E}}\big) = \frac{1}{r}\sum_{i=1}^{r}\arccos\left(\frac{\langle \mathbf{e}_{(i)}, \hat{\mathbf{E}}_{(i)}\rangle}{\|\mathbf{e}_{(i)}\|_2\|\hat{\mathbf{E}}_{(i)}\|_2}\right)\frac{180}{\pi} \quad (12)$$

where $\langle\cdot\rangle$ denotes the inner product and $\mathbf{e}_{(i)}$ indicates the ith column of $\mathbf{E}$.

C. Unmixing Experiments

1) Experiments on Simulated Datasets: To evaluate the robustness of the methods to noise, the Gaussian white noise with varying noise power is added to both simulated datasets to obtain data with SNR of 20, 30, 40, and 50 dB. Table II shows the results in terms of RMSE (in percentage) for the simulated dataset 1. CyCUNet produced the worst results. FCLSU and Collab both show poor performances. UnDIP outperforms uDAS, but its results are not competitive with NMF-QMV and MiSiCNet. Both NMF-QMV and MiSiCNet considerably outperform the other techniques in terms of RMSE. With higher amounts of noise, however, MiSiCNet gradually improves over NMF-QMV.

Table III reports the RMSE results for the Simulated Dataset 2. The trends are similar; however, all the competing methods perform poorly. MiSiCNet shows considerable improvements even compared to NMF-QMV. In addition, the very low standard deviations reported in the tables confirm the robustness of MiSiCNet for different noise levels. That is a valuable advantage revealed from the experiments. For instance, the small (0.09%) performance improvement of NMF-QMV over MiSiCNet for the simulated dataset 1 with 20 dB of noise is statistically insignificant due to the high standard deviation of 0.2% for NMF-QMV compared to the very low standard deviation of 0.07% for MiSiCNet. Overall, NMF-QMV and MiSiCNet outperform the other techniques in terms of RMSE. However, MiSiCNet outperforms NMF-QMV for high SNRs. This could be attributed to the proposed parameter selection technique for the geometrical penalty (β) in NMF-QMV. In addition, the search interval ($\beta \in 10^{-5}, 10^{-4}, \ldots, 10^{4}, 10^{5}$) might not be suitable for all datasets with varying SNRs.

Tables IV and V report SAD for simulated datasets 1 and 2, respectively. The results follow the trend of RMSE. MiSiCNet considerably outperforms the other techniques for all SNRs for Simulated dataset 2. For simulated dataset 1, NMF-QMV slightly outperforms MiSiCNet for 20 and 30 dB. As we already discussed, this could be attributed to the parameter selection technique in NMF-QMV. The search interval is too large to pick the optimum parameter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

TABLE III
RMSE (SIMULATED 2). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
20dB	12.55±1.89	12.15±1.04	11.43±2.69	19.51±5.89	12.05±0.61	4.03±0.54	**3.96±0.04**
30dB	21.45±2.49	10.49±0.21	12.57±5.11	15.87±2.71	13.83±1.94	3.79±2.33	**2.45±0.02**
40dB	21.6±4.11	10.52±0.22	10.84±4.29	14.57±1.3	14.11±1.93	7.37±1.13	**2.15±0.03**
50dB	22.89±2.71	10.37±0.17	10.76±4.24	15.96±2.02	14.14±1.18	6.91±1.17	**2.12±0.03**

TABLE IV
SAD (SIMULATED 1). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA	SiVM	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
20dB	4.64±0.85	6.55±0.6	7.72±4.43	8.37±0.89	6.37±0.81	**2.06±0.25**	2.53±0.64
30dB	4.53±0.96	6.76±0.07	7.33±5.17	8.59±1.03	6.21±1.26	**0.56±0.22**	0.95±0.1
40dB	4.26±0.57	6.84±0.06	7.75±3.74	9.32±0.63	5.67±0.82	0.83±0.51	**0.63±0.09**
50dB	4.66±0.86	6.85±0.07	6.68±3.62	9.83±0.52	5.73±1.31	1.84±0.5	**0.69±0.11**

TABLE V
SAD (SIMULATED 2). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA	SiVM	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
20dB	7.83±0.88	8.03±0.08	11.77±3.62	9.41±0.63	8.36±0.63	2.47±0.48	**1.76±0.03**
30dB	7.72±1.29	7.83±0.02	14.68±4.42	9.73±0.5	7.21±0.57	8.45±7.73	**0.83±0.02**
40dB	8.24±1.02	7.85±0.01	13.95±5.97	9.96±0.6	7.82±1.2	22.95±2.75	**0.64±0.02**
50dB	7.73±1.23	7.86±0.01	14.47±5.77	10.57±0.23	7.65±0.28	24.13±1.26	**0.62±0.02**

TABLE VI
RMSE (SAMSON DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Soil	17.66	17.78	17.99	24.17	15.06	52.35	**2.76**
Tree	6.53	13.30	13.83	13.86	6.07	39.90	**2.44**
Water	14.92	20.96	23.03	26.54	11.81	45.98	**2.07**
Overall	13.87	17.63	18.67	22.21	11.59	46.36	**2.44**

TABLE VII
SAD (SAMSON DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA	SiVM	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Soil	1.49	1.49	2.05	6.55	0.89	2.24	**0.64**
Tree	5.51	4.28	5.50	8.69	4.77	7.10	**2.60**
Water	8.91	8.91	8.75	11.92	**8.03**	87.10	21.27
Overall	5.30	4.89	5.43	9.06	**4.56**	32.14	8.17

Both simulated experiments reveal the importance of the minimum volume penalty term in the absence of pure pixels. Both NMF-QMV and MisiCNet consider the geometry of the data simplex by minimizing the volume term, while all the other techniques either ignore that or rely on the presence of pure pixels, for instance, by using VCA (in FCLSU) and SiVM (in UnDIP).

Figs. 8 and 9 depict the abundances and endmembers estimated by the different unmixing techniques on the simulated dataset 2 (40 dB). Visual comparisons confirm the advantage of MiSiCNet compared to the other methods. MiSiCNet provides excellent abundance estimations for Endmembers 1 and 6, and good estimations for the other ones. NMF-QMV also outperforms the rest of the methods but fails to estimate satisfactory abundances for this challenging mixing scenario. From Fig. 9, we can see that MiSiCNet estimates all endmembers successfully, except for a slight mismatch in Endmember 5. On the other hand, all the other techniques perform poorly for all the cases.

2) Experiments on Real Datasets: Table VI shows the abundance RMSE obtained by the different unmixing techniques to the Samson data. MiSiCNet significantly improved the abundance estimation for this dataset compared to the other methods. In addition, MiSiCNet considerably outperformed the other techniques for all three individual abundances. MiSiCNet performed 12.3%, 3.63%, and 9.74% better on the soil, tree, and water abundances, respectively, compared with the second-best results, given by Collab. Figs. 10 and 11 depict the estimated abundances and endmembers, respectively, obtained by the different techniques. This again confirms that MiSiCNet outperforms the other techniques. Table VII reports the SAD values of the endmembers for Samson. MiSiCNet outperforms the other techniques in terms of SAD for the soil and tree abundances but not for the water abundances. Overall, Collab shows 3.61° SAD improvement over MiSiCNet. SAD does not follow the trend of the abundance RMSE in real datasets, an effect that is caused by spectral variability. As SAD removes the norm of the endmember spectra, it ignores endmember scaling factors (caused by multiple reflections of the light and variable illumination conditions). However, such scaling factors may considerably affect the abundance estimation. Tables VIII and IX compare the abundance and endmember estimation on the Apex data of the different unmixing techniques in terms of RMSE and SAD. Figs. 12 and 13 show the estimated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

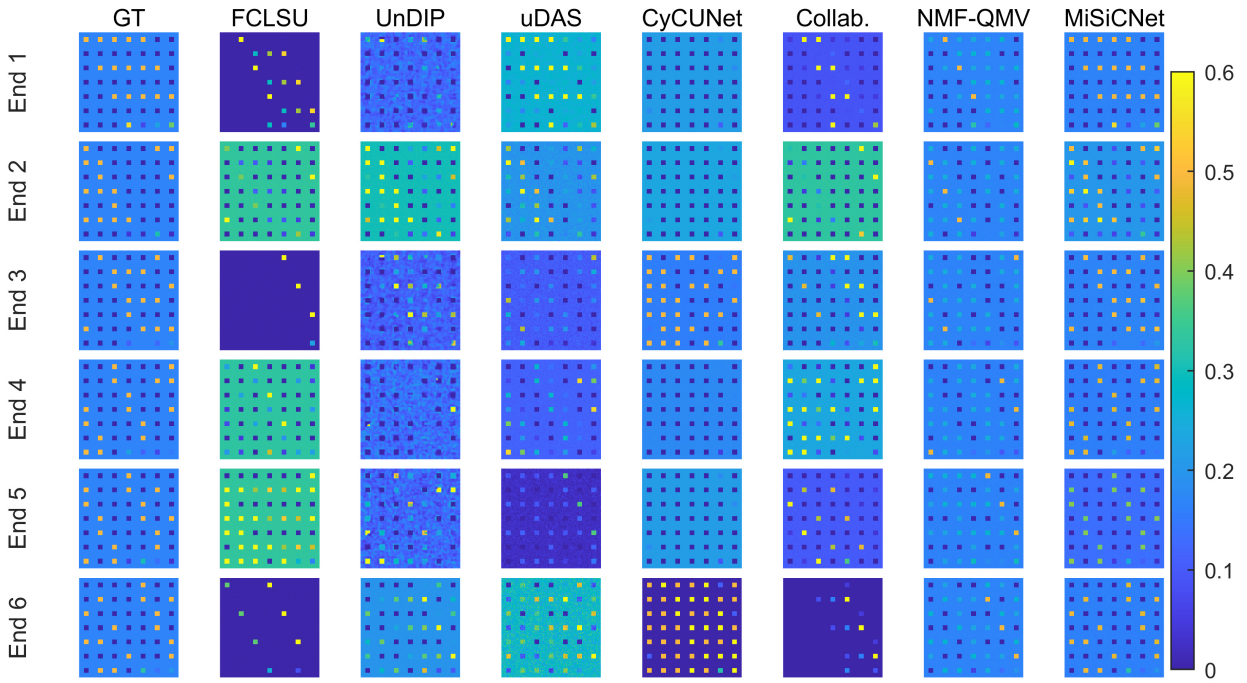


Fig. 8. Simulated dataset 2 (40 dB). The visual comparison of the abundance maps obtained by applying different unmixing techniques.

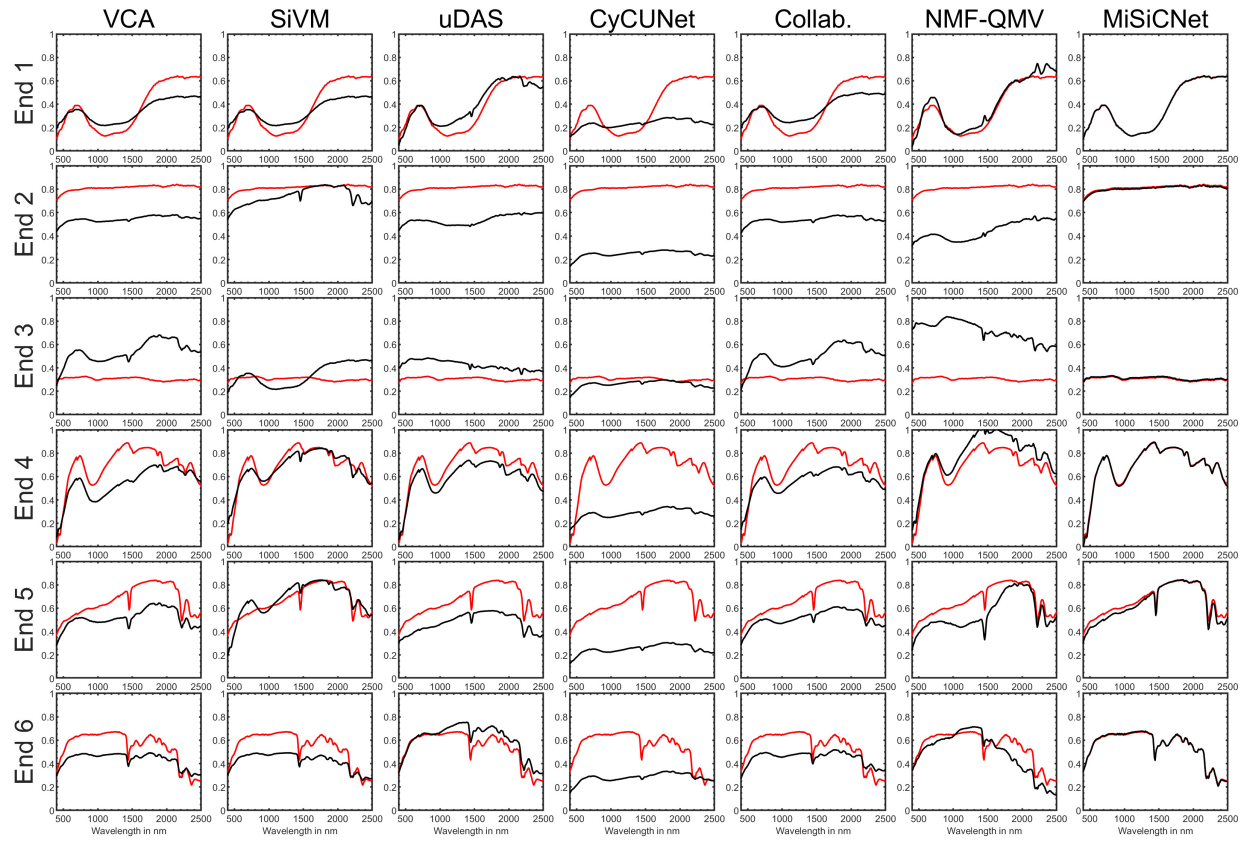


Fig. 9. Simulated dataset 2 (40 dB). The visual comparison of the endmembers obtained by applying different unmixing techniques. Red: ground-truth endmembers. Black: estimated endmembers.

abundances and endmembers, respectively. From Table VIII, we can see that MiSiCNet slightly outperforms FCLSU, and they both improve the abundance estimation of Apex with 5% RMSE over the next best results. SiVM obtained the best results in terms of SAD. The effect of neglecting the endmember scaling factor by SAD is much clearer in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

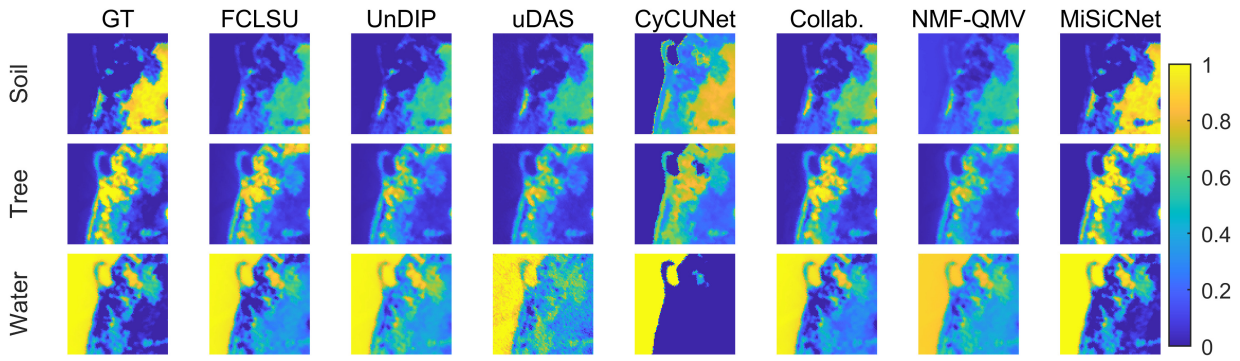


Fig. 10. Samson dataset. The visual comparison of the abundance maps obtained by the different unmixing techniques.

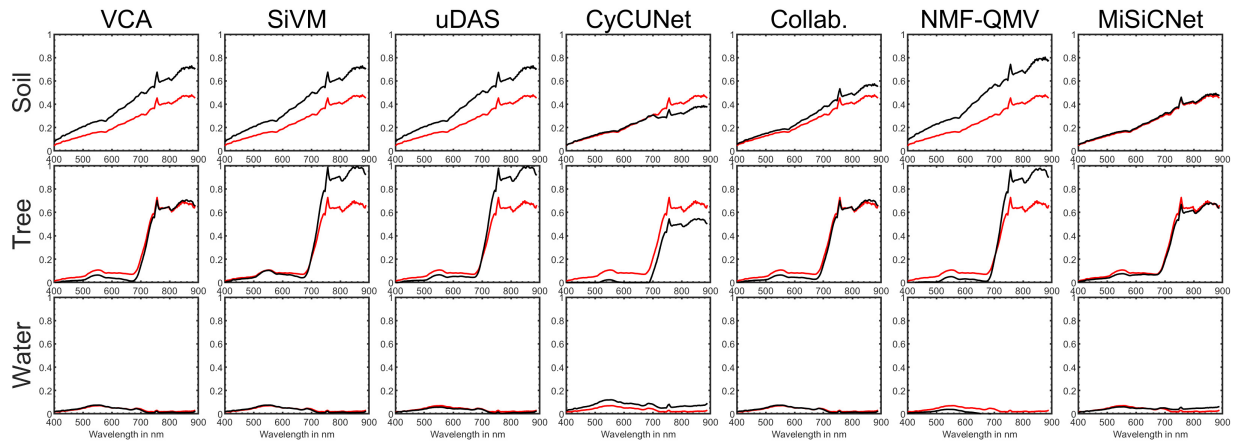


Fig. 11. Samson dataset. The visual comparison of the endmembers obtained by the different unmixing techniques. Red: ground-truth endmembers. Black: estimated endmembers.

TABLE VIII

RMSE (APEX DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Road	23.28	17.42	19.81	29.38	30.83	18.14	**13.67**
Tree	**9.46**	21.54	14.28	20.15	18.92	24.69	14.92
Roof	12.06	26.78	24.27	16.78	15.54	24.87	**11.38**
Water	13.10	42.64	30.01	12.01	**7.90**	38.41	19.84
Overall	15.40	28.74	22.84	20.58	20.08	27.54	**15.27**

TABLE IX

SAD (APEX DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA	SiVM	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Road	39.62	**5.20**	26.07	26.03	38.80	22.94	14.70
Tree	15.15	7.67	8.05	**4.87**	11.82	15.53	7.91
Roof	8.43	**3.94**	4.92	7.43	5.74	10.04	10.05
Water	29.66	28.88	**12.90**	35.66	29.43	74.48	43.43
Overall	23.21	**11.42**	12.99	18.50	21.45	30.75	19.02

the results of Apex. Comparing the estimated endmembers by SiVM and MiSiCNet in Fig. 13 reveals the difference between the two estimations. However, the reported results for SAD in Table IX show that SiVM obtained the best results on the road endmember, again because SAD neglects scaling factors. On the other hand, Fig. 12 confirms that MiSiCNet provided the best road abundance estimation. Overall, the results confirm the advantage of the MiSiCNet over the other unmixing techniques.

Tables X and XI show the abundance RMSE and SAD, respectively, for the Washington DC Mall dataset. On this dataset, MiSiCNet improved the overall abundance RMSE and SAD by 5.69% and 1.3° over the second-best results provided by NMF-QMV and Collab, respectively. In addition, MiSiCNet improved the abundance estimation of grass, tree, road, and water by 6%, 17.51%, 2.11%, and 4.53%, respectively, over the second-best results. MiSiCNet did not improve the abundance estimations for trail and roof for which NMF-QMV and FCLSU, respectively, obtained better abundance estimations. The estimated abundances and endmembers on the Washington DC Mall dataset are shown in Figs. 14 and 15, respectively. The visual comparison confirms the reported results for the abundance RMSE.

D. Sensitivity Analysis to Hyperparameters

In this section, we discuss the sensitivity of MiSiCNet to the selection of hyperparameters. We selected the four most important hyperparameters that can affect the performance of MiSiCNet: the regularization parameter, the LR, the number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

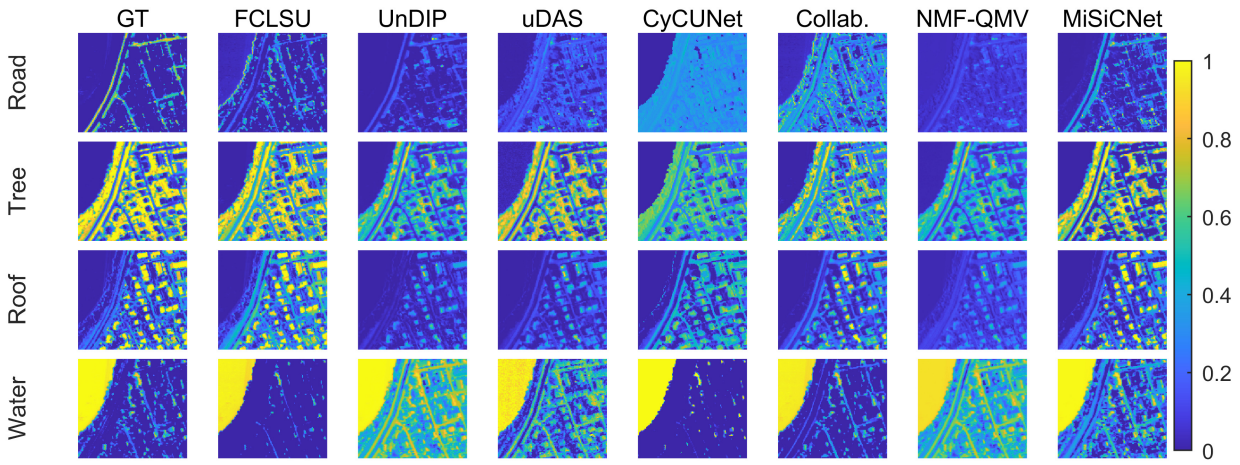


Fig. 12. Apex dataset. The visual comparison of the abundance maps obtained by the different unmixing techniques.

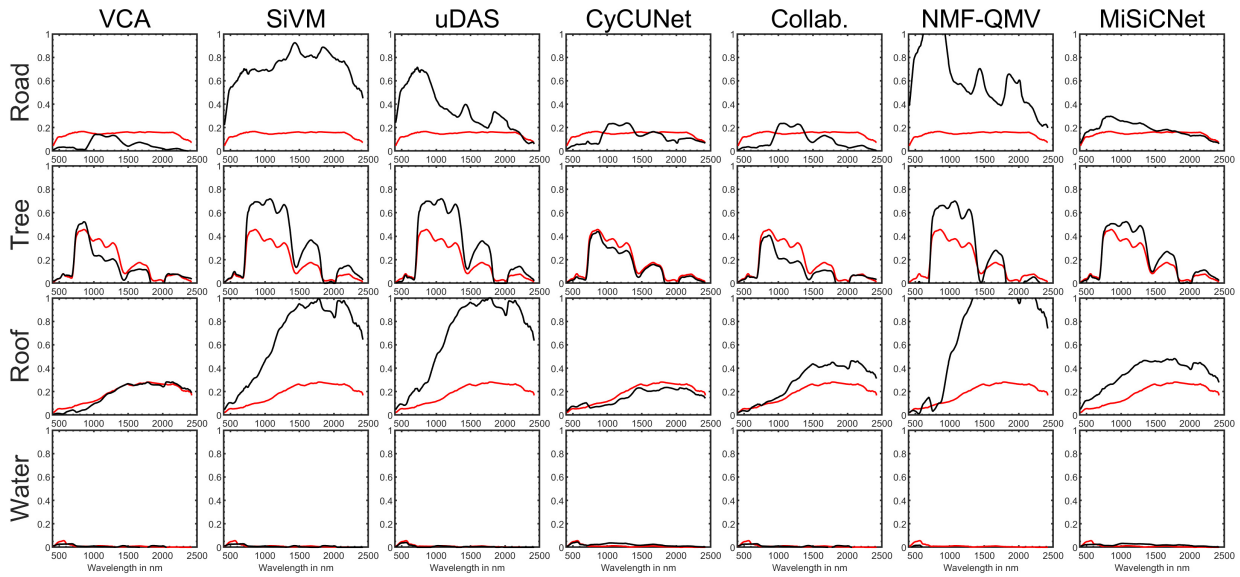


Fig. 13. Apex dataset. The visual comparison of the endmembers obtained by the different unmixing techniques. Red: ground-truth endmembers. Black: estimated endmembers.

TABLE X

RMSE (WASHINGTON DC MALL DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Grass	30.18	29.09	37.34	40.20	28.38	35.81	**22.38**
Tree	39.73	34.93	33.11	28.04	41.26	27.30	**9.79**
Road	17.55	24.24	24.80	25.28	22.58	23.51	**15.44**
Roof	**3.82**	4.88	4.65	40.89	4.48	8.40	16.76
Water	29.07	38.02	51.36	39.86	30.91	20.97	**16.44**
Trail	12.49	23.72	18.26	21.32	19.34	**10.30**	20.67
Overall	25.21	27.95	31.88	33.56	26.98	23.08	**17.39**

TABLE XI

SAD (WASHINGTON DC MALL DATASET). THE BEST PERFORMANCES ARE SHOWN IN BOLD

	VCA	SiVM	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Grass	18.16	10.60	10.87	**5.13**	18.17	11.18	16.68
Tree	16.52	41.58	24.35	15.49	19.11	25.83	**9.48**
Road	13.27	49.32	37.73	26.60	19.71	12.85	**3.37**
Roof	1.97	16.19	11.41	54.43	**1.90**	11.91	19.11
Water	44.50	54.40	13.34	24.09	**1.75**	38.60	3.80
Trail	37.08	10.05	5.39	45.30	19.74	**3.52**	20.14
Overall	21.92	30.36	17.18	28.51	13.40	17.31	**12.10**

of filters, and the kernel size. Mean values of RMSE and SAD are shown, along with the standard deviations as error bars.

1) Regularization Parameter: Fig. 16(a) shows SAD and RMSE values from the estimated endmembers and abundances, respectively, when varying the tuning parameter λ from $\{10^{-3}, 10^{-2}, \ldots, 10^{3}\}$. The best results were obtained for $\lambda = 10^{-1}$. Note that SAD by itself (applied on the endmembers) is not sufficient to evaluate the performance. However, when combined with a quality metric, such as the abundance RMSE, the performance of the proposed technique

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

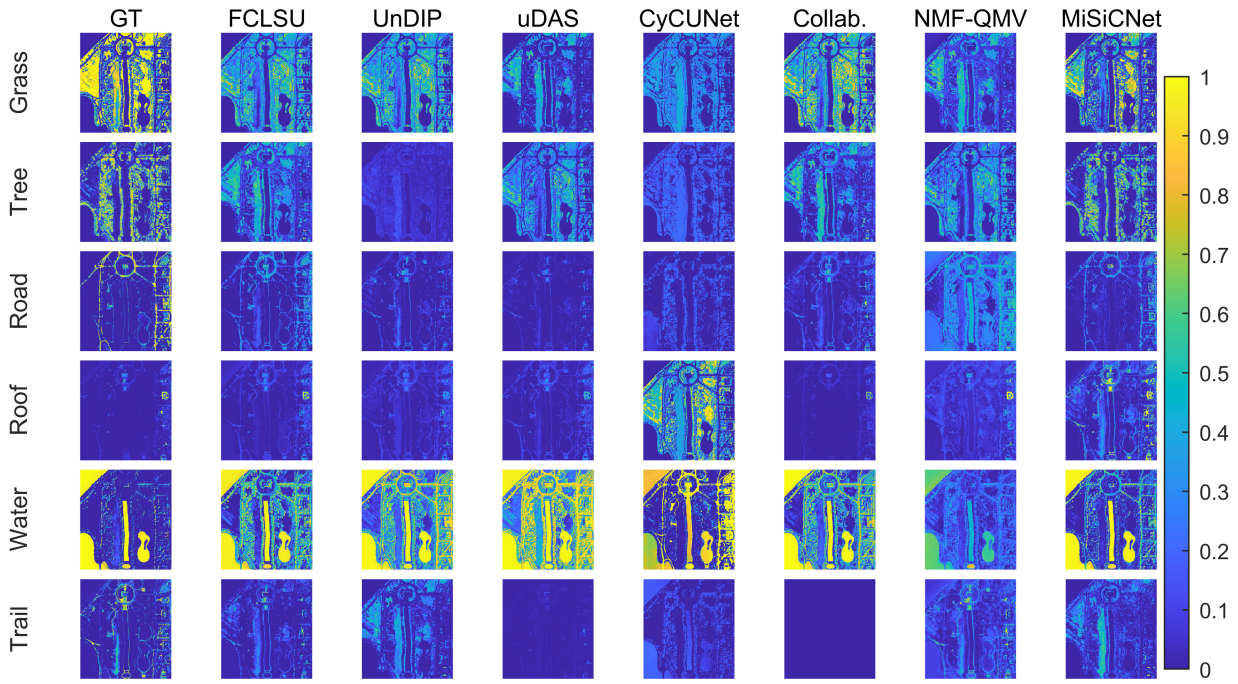


Fig. 14. Washington DC Mall dataset. The visual comparison of the abundance maps obtained by the different unmixing techniques.

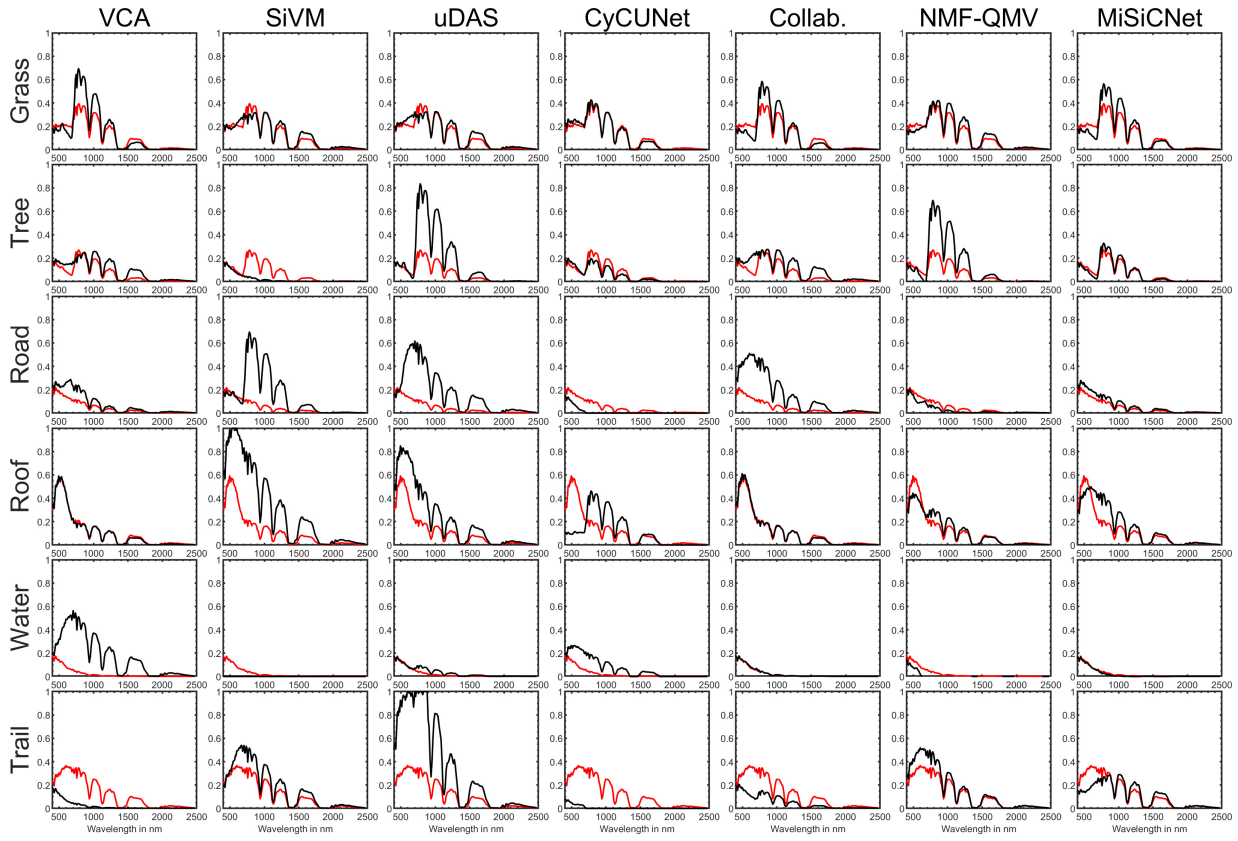


Fig. 15. Washington DC Mall. The visual comparison of the endmembers obtained by applying the different unmixing techniques. Red: ground-truth endmembers. Black: estimated endmembers.

is revealed. Therefore, the low value of SAD at 10^3 along with a high abundance RMSE does not show a good performance. We should note that λ is the only parameter that needs to be set for different datasets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

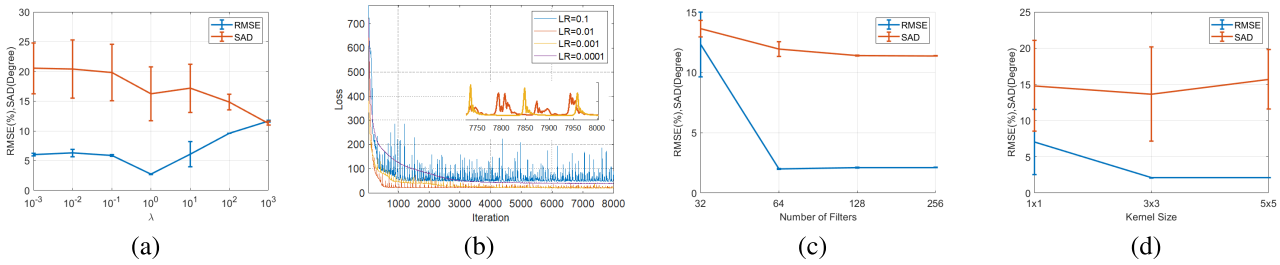


Fig. 16. Sensitivity of MiSiCNet to the hyperparameters of the network. The experiments were performed on the simulated dataset 2 (50 dB). (a) Regularization parameter. (b) Learning rate. (c) Number of filters. (d) Kernel size.

2) Learning Rate: Fig. 16(b) shows the loss function values for different LRs, i.e., 0.1, 0.01, 0.001, and 0.0001. For LR $=$ 0.1, the loss function value shows large variations. LR $=$ 0.0001 shows less variation, but the loss function converges to a higher minimum value. Both LR $=$ 0.01 and 0.001 perform similarly, but LR $=$ 0.001 shows fewer variations, and the minimum loss is slightly lower than for LR $=$ 0.01. Therefore, we suggest using LR $=$ 0.001 for the proposed method.

3) Number of Filters: Fig. 16(c) shows the abundance RMSE and SAD with respect to the number of filters used for the convolutional layers. It can be observed that both abundance RMSE and SAD suggest the selection of 256 filters.

4) Kernel Size: Fig. 16(d) shows the abundance RMSE and SAD with respect to the size of the kernels used in the convolutional layers. Both SAD and abundance RMSE give the lowest values for filters of size 3×3, confirming our selection.

E. Robustness w.r.t. the Endmember Initialization

A significant challenge in blind unmixing techniques is the initialization of the endmembers. Problem (2) is a nonconvex optimization problem due to the nonconvex fidelity term, and therefore, the initialization plays a significant role in the performance of the final algorithm. In blind unmixing techniques, this is often tackled by using geometrical approaches, such as VCA and SiVM. However, those approaches cannot tackle datasets without pure pixels or even more difficult scenarios, such as those considered in this article. As a result, the performance of blind unmixing techniques is often not satisfactory due to poor initialization.

On the other hand, deep learning-based methods can cope with the initialization challenge since they search for optimum weights in the high-dimensional space. Often, the minimum becomes a settled point in that space, which can be reached by optimization algorithms such as gradient descent. We should note that the endmembers are the weights of the decoder and can also be initialized randomly, such as the other weights. Fig. 17 shows the performance of MiSiCNet in terms of RMSE and SAD applied to the simulated dataset 2 (50 dB) using three different initialization strategies of the weights of the decoder.

1) *Random Vectors:* Initializing randomly in the same way as the other layers.
2) *Random Pixels:* Initializing using a random selection of spectral pixels.
3) Initializing using the endmembers extracted by SiVM.

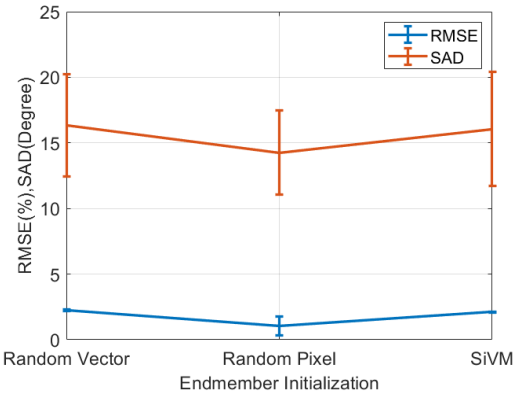


Fig. 17. Comparison of different strategies for initializing the decoder weights in MiSiCNet. The performance is given in terms of RMSE in percentage.

All the results are mean values over five experiments. The best result is given by random pixels with around 1% improvement in terms of abundance RMSE and 1.8° improvement in terms of SAD compared to the SiVM initialization. However, the standard deviation for the abundance RMSE is much higher (0.71% compared to 0.02%) because of the random selection of the pixels.

On the other hand, SiVM shows robustness to the random noise, and hence, we selected it for the initialization of MiSiCNet. In addition, the results confirm the robustness of MiSiCNet w.r.t. that initialization, and thus, the performance of MiSiCNet does not rely on an endmember extraction technique and, therefore, is successful in the absence of pure pixels. We should note that, for random vectors, we used LR $=$ 0.01 to have a faster convergence.

F. Processing Time

Table XII gives the processing times for the different unmixing techniques applied to the three real datasets. FCLSU, uDAS, Collab, and NMF-QMV were implemented in MATLAB (2020b). UnDIP, CyCUNet, and MiSiCNet were implemented in Python (3.8). The reported processing times are mean values over five experiments and were obtained using a computer with an Intel[2] Core[3] i9-10980 HK processor (2.4 GHz), 32 GB of memory, a 64-bit Operating System,

[2]Registered trademark.
[3]Trademarked.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

TABLE XII
PROCESSING TIME (IN SECONDS) OF THE UNMIXING TECHNIQUES APPLIED TO THE REAL DATASETS

	FCLSU	UnDIP	uDAS	CyCUNet	Collab.	NMF-QMV	MiSiCNet
Samson	**1.05**	48.19	19.57	69.21	11.45	9.65	90.11
Apex	**1.56**	87.33	262.49	128.86	25.35	16.45	135.72
WDC	**12.9**	431.95	1.05e+03	854.73	221.6	491.26	727.04

and an NVIDIA GEFORCE RTX (2080 Super) graphical processing unit. The table shows that conventional approaches are generally faster than DL-based ones, and FCLSU is the most rapid unmixing technique. However, due to the efficiency of GPU programming, the processing time of MiSiCNet is acceptable, and its use can be justified by the significant performance improvements. It is worth mentioning that MiSiCNet is less affected by the increase of the spatial size from Samson to WDC than the other techniques.

G. Discussion

FCLSU and UnDIP estimate abundances using endmember extraction techniques, i.e., VCA and SiVM. Therefore, their performances depend on the performance of the endmember extraction techniques, which are often designed to assume pure pixels. Thus, in the absence of pure pixels, such methods provide poor performances. uDAS is a DAEN, which neither incorporates the spatial information nor the geometrical information, and therefore, its overall performance is not satisfactory. CyCUNet includes spatial information by using convolutional autoencoders, but it ignores the geometrical information. In addition, its poor performance can be attributed to the absence of ASC during the optimization process. CyCUNet applies the Clamp function instead of Softmax on the abundances for the optimization, and ASC needs to be enforced after estimating the abundances. Collab uses a sparse regression to estimate the abundances using a library of endmembers created from the dataset. Hence, Collab cannot perform successfully in the absence of pure pixels. NMF-QMV incorporates geometrical information, but it does not include spatial information. In addition, the searching range for the parameter selection is too extensive such that an automatic selection technique may miss the optimal parameter values. We should note that NMF-QMV solves the unmixing problem in a subspace, i.e., it assumes that the endmembers live in an $r-1$ dimensional affine set, which best represents the data using an orthogonal projection. Such an assumption helps to remove the outliers and noise. However, the constraint on the endmembers cannot be enforced throughout the optimization, and therefore, the endmembers cannot be bounded between zero and one. Accordingly, negative values for the endmembers may be estimated and should be set to zero at the end of the optimization.

On the other hand, MiSiCNet applies all the constraints in (1) throughout the optimization. More importantly, MiSiCNet incorporates spatial information by both using convolutional filters and implicitly applying a regularizer on the abundances. In addition, geometrical information is incorporated using a penalty term. We should note that the tuning parameter λ was not optimally tuned but fixed to 100 for all the real datasets for a fair comparison with the other techniques.

V. CONCLUSION

We proposed the MiSiCNet for deep hyperspectral unmixing. We have shown that a deep convolutional network can solve a blind unmixing problem defined as a constrained penalized least-squares optimization. Indeed, we show that such an optimization can be shifted toward an optimization on the deep network's parameters. We evaluated the performance of the proposed approach using two simulated and three real datasets. The simulated datasets consider realistic and challenging scenarios when there are no pure pixels and only one or two data points on the facets of the data simplex. The results on both the simulated and real datasets show considerable improvements over the state of the art. In addition, the experimental results reveal the importance of the simplex volume penalty term for unmixing when no pure pixels are available.

REFERENCES

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] P. Ghamisi *et al.*, "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[4] N. Dobigeon, J.-Y. Tourneret, C. Richard, J. C. M. Bermudez, S. McLaughlin, and A. O. Hero, "Nonlinear unmixing of hyperspectral images: Models and algorithms," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 82–94, Jan. 2014.

[5] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[6] R. Heylen, D. Burazerovic, and P. Scheunders, "Fully constrained least squares spectral unmixing by simplex projection," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4112–4122, Nov. 2011.

[7] T.-H. Chan, C.-Y. Chi, Y.-M. Huang, and W.-K. Ma, "A convex analysis-based minimum-volume enclosing simplex algorithm for hyperspectral unmixing," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4418–4432, Nov. 2009.

[8] J. Nascimento and J. Bioucas-Dias, "Vertex component analysis: A fast algorithm to extract endmembers spectra from hyperspectral data," in *Pattern Recognition and Image Analysis*, F. J. Perales, A. J. C. Campilho, N. P. de la Blanca, and A. Sanfeliu, Eds. Berlin, Germany: Springer, 2003, pp. 626–635.

[9] J. Boardman, F. A. Kruse, and R. Green, "Mapping target signatures via partial unmixing of AVIRIS data," in *Proc. JPL Airborne Earth Sci. Workshop*, 1995, pp. 23–26.

[10] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[11] W. E. Full, R. Ehrlich, and J. E. Klovan, "EXTENDED QMODEL—Objective definition of external end members in the analysis of mixtures," *J. Int. Assoc. Math. Geol.*, vol. 13, no. 4, pp. 331–344, Aug. 1981.

[12] M. D. Craig, "Minimum-volume transforms for remotely sensed data," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 3, pp. 542–552, May 1994.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.

[13] C.-H. Lin, R. Wu, W.-K. Ma, C.-Y. Chi, and Y. Wang, "Maximum volume inscribed ellipsoid: A new simplex-structured matrix factorization framework via facet enumeration and convex optimization," *SIAM J. Imag. Sci.*, vol. 11, no. 2, pp. 1651–1679, Jan. 2018.

[14] J. Li, J. M. Bioucas-Dias, and A. Plaza, "Collaborative nonnegative matrix factorization for remotely sensed hyperspectral unmixing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2012, pp. 3078–3081.

[15] N. Dobigeon, S. Moussaoui, M. Coulon, J.-Y. Tourneret, and A. O. Hero, "Joint Bayesian endmember extraction and linear unmixing for hyperspectral imagery," *IEEE Trans. Signal Process.*, vol. 57, no. 11, pp. 4355–4368, Nov. 2009.

[16] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Blind hyperspectral unmixing using total variation and ℓ_q sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6371–6384, Nov. 2016.

[17] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[18] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.

[19] M. Berman, H. Kiiveri, R. Lagerstrom, A. Ernst, R. Dunne, and J. F. Huntington, "ICE: A statistical approach to identifying endmembers in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 10, pp. 2085–2095, Oct. 2004.

[20] L. Zhuang, C. Lin, M. A. T. Figueiredo, and J. M. Bioucas-Dias, "Regularization parameter selection in minimum volume hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9858–9877, Dec. 2019.

[21] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 3, Jul. 2008, pp. III-250–III-253.

[22] J. M. Bioucas-Dias, "A variable splitting augmented Lagrangian approach to linear spectral unmixing," in *Proc. 1st Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Aug. 2009, pp. 1–4.

[23] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Aug. 2011.

[24] M. D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[25] B. Rasti and B. Koirala, "SUnCNN: Sparse unmixing using unsupervised convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] L. Meier, S. van de Geer, and P. Bühlmann, "The group lasso for logistic regression," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 70, no. 1, pp. 53–71, Feb. 2008.

[27] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[28] L. Drumetz, T. R. Meyer, J. Chanussot, A. L. Bertozzi, and C. Jutten, "Hyperspectral image unmixing with endmember bundles and group sparsity inducing mixed norms," *IEEE Trans. Image Process.*, vol. 28, no. 7, pp. 3435–3450, Jul. 2019.

[29] B. Rasti *et al.*, "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[30] Y. Su, A. Marinoni, J. Li, J. Plaza, and P. Gamba, "Stacked nonnegative sparse autoencoders for robust hyperspectral unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1427–1431, Sep. 2018.

[31] Y. Su, J. Li, A. Plaza, A. Marinoni, P. Gamba, and S. Chakravortty, "DAEN: Deep autoencoder networks for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4309–4321, Jul. 2019.

[32] R. A. Borsoi, T. Imbiriba, and J. C. M. Bermudez, "Deep generative endmember modeling: An application to unsupervised spectral unmixing," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 374–384, 2020.

[33] Y. Qu and H. Qi, "uDAS: An untied denoising autoencoder with sparsity for spectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1698–1712, Mar. 2019.

[34] S. Ozkan, B. Kaya, and G. B. Akar, "EndNet: Sparse AutoEncoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[35] B. Palsson, J. Sigurdsson, J. R. Sveinsson, and M. O. Ulfarsson, "Hyperspectral unmixing using a neural network autoencoder," *IEEE Access*, vol. 6, pp. 25646–25656, 2018.

[36] Z. Hua, X. Li, Q. Qiu, and L. Zhao, "Autoencoder network for hyperspectral unmixing with adaptive abundance smoothing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 9, pp. 1640–1644, Sep. 2021.

[37] X. Zhang, Y. Sun, J. Zhang, P. Wu, and L. Jiao, "Hyperspectral unmixing via deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 11, pp. 1755–1759, Nov. 2018.

[38] B. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Spectral–spatial hyperspectral unmixing using multitask learning," *IEEE Access*, vol. 7, pp. 148861–148872, 2019.

[39] F. Khajehrayeni and H. Ghassemian, "Hyperspectral unmixing using deep convolutional autoencoders in a supervised scenario," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 567–576, 2020.

[40] B. Palsson, M. O. Ulfarsson, and J. R. Sveinsson, "Convolutional autoencoder for spectral–spatial hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 535–549, Jan. 2021.

[41] B. Rasti, B. Koirala, P. Scheunders, and P. Ghamisi, "UnDIP: Hyperspectral unmixing using deep image prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[42] V. Lempitsky, A. Vedaldi, and D. Ulyanov, "Deep image prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 9446–9454.

[43] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Deep image prior," *Int. J. Comput. Vis.*, vol. 128, no. 7, pp. 1867–1888, Mar. 2020.

[44] L. Gao, Z. Han, D. Hong, B. Zhang, and J. Chanussot, "CyCU-Net: Cycle-consistency unmixing network by learning cascaded autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[45] C.-H. Lin, W.-K. Ma, W.-C. Li, C.-Y. Chi, and A. Ambikapathi, "Identifiability of the simplex volume minimization criterion for blind hyperspectral unmixing: The no-pure-pixel case," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 10, pp. 5530–5546, Oct. 2015.

[46] I. J. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016. [Online]. Available: http://www.deeplearningbook.org

[47] F. Zhu, Y. Wang, B. Fan, S. Xiang, G. Meng, and C. Pan, "Spectral unmixing via data-guided sparsity," *IEEE Trans. Image Process.*, vol. 23, no. 12, pp. 5412–5427, Dec. 2014.

[48] M. E. Schaepman *et al.*, "Advanced radiometry measurements and Earth science applications with the airborne prism experiment (APEX)," *Remote Sens. Environ.*, vol. 158, pp. 207–219, Mar. 2015.



Behnood Rasti (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees both in electronics and electrical engineering from the Electrical Engineering Department, University of Guilan, Rasht, Iran, in 2006 and 2009, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Iceland, Reykjavik, Iceland, in 2014.

From 2015 to 2016, he was a Post-Doctoral Researcher with the Electrical and Computer Engineering Department, University of Iceland. From 2016 to 2019, he was a Lecturer with the Center of Engineering Technology and Applied Sciences, Department of Electrical and Computer Engineering, University of Iceland. He was a Humboldt Research Fellow in 2020 and 2021. He is currently a Principal Research Associate with Helmholtz-Zentrum Dresden-Rossendorf (HZDR), Freiberg, Germany. His research interests include signal and image processing, machine/deep learning, remote sensing, and artificial intelligence.

Dr. Rasti was the Valedictorian as an M.Sc. student in 2009. He won the Doctoral Grant of The University of Iceland Research Fund "The Eimskip University Fund" and the "Alexander von Humboldt Research Fellowship Grant" in 2013 and 2019, respectively. He also serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS (GRSL).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.



Bikram Koirala (Member, IEEE) received the M.S. degree in geomatics engineering from the University of Stuttgart, Stuttgart, Germany, in 2016, and the Ph.D. degree in development of advanced hyperspectral unmixing methods from the University of Antwerp, Antwerp, Belgium, in 2021.

In 2017, he joined the Vision Lab, Department of Physics, University of Antwerp, as a Ph.D. Researcher, where he is currently a Post-Doctoral Researcher. His research interests include machine learning and hyperspectral image processing.



Paul Scheunders (Senior Member, IEEE) received the B.S. and Ph.D. degrees in physics with a focus on statistical mechanics from the University of Antwerp, Antwerp, Belgium, in 1983 and 1990, respectively.

In 1991, he became a Research Associate at the Vision Lab, Department of Physics, University of Antwerp, where he is currently a Full Professor. He has published over 200 papers in international journals and proceedings in the field of image processing, pattern recognition, and remote sensing. His research interests include remote sensing and hyperspectral image processing.

Dr. Scheunders has served as a program committee member in numerous international conferences. He is also a Senior Member of the IEEE Geoscience and Remote Sensing Society. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Jocelyn Chanussot (Fellow, IEEE) received the M.Sc. degree in electrical engineering from the Grenoble Institute of Technology (Grenoble INP), Grenoble, France, in 1995, and the Ph.D. degree from the Université de Savoie, Annecy, France, in 1998.

Since 1999, he has been with Grenoble INP, where he is currently a Professor of signal and image processing. He has been a Visiting Scholar with Stanford University, Stanford, CA, USA, the KTH Royal Institute of Technology, Sweden, and the National University of Singapore (NUS), Singapore. Since 2013, he has been an Adjunct Professor with the University of Iceland, Reykjavik, Iceland. From 2015 to 2017, he was a Visiting Professor with the University of California at Los Angeles (UCLA), Los Angeles, CA, USA. He holds the AXA Chair in remote sensing and is an Adjunct Professor with the Chinese Academy of Sciences, Aerospace Information Research Institute, Beijing, China. His research interests include image analysis, hyperspectral remote sensing, data fusion, machine learning, and artificial intelligence.

Dr. Chanussot was a member of the Machine Learning for Signal Processing Technical Committee of the IEEE Signal Processing Society from 2006 to 2008. He is also a member of the Institut Universitaire de France from 2012 to 2017. He has been a Highly Cited Researcher (Clarivate Analytics/Thomson Reuters) since 2018. He received multiple outstanding paper awards. He was the General Chair of the first IEEE GRSS Workshop on Hyperspectral Image and Signal Processing, Evolution in Remote Sensing (WHISPERS). He was the Chair and the Co-Chair of the GRS Data Fusion Technical Committee from 2009 to 2011 and 2005 to 2008, respectively. He was the Program Chair of the IEEE International Workshop on Machine Learning for Signal Processing in 2009. He was the Vice-President of the IEEE Geoscience and Remote Sensing Society, in charge of meetings and symposia from 2017 to 2019. He was the Founding President of the IEEE Geoscience and Remote Sensing French Chapter from 2007 to 2010, which received the 2010 IEEE GRS-S Chapter Excellence Award. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE TRANSACTIONS ON IMAGE PROCESSING, and the PROCEEDINGS OF THE IEEE. He was the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING from 2011 to 2015. In 2014, he has served as a Guest Editor for the *IEEE Signal Processing Magazine*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:46:49 UTC from IEEE Xplore. Restrictions apply.